

Cole Bregman

(650) 229-2700 | ctb2159@columbia.edu | [linkedin.com/in/cole-bregman](https://www.linkedin.com/in/cole-bregman) | [colebregman.com](https://www.colebregman.com)

Education

Columbia University

New York, NY · 2023 – 2027

B.S. Mechanical Engineering, Minor in Computer Science | GPA: 4.0 | Dean's list | Tau Beta Pi

Work Experience

Walt Disney Imagineering

Glendale, CA

Ride Development Engineering Intern

June 2026 - August 2026

- Designed station gates for upcoming attractions using solidworks, matching design intent and safety standards.
- Conducted FEA, FMEA and hazard analysis for station gate systems & vehicles on upcoming attractions.

Apple Inc.

Cupertino, CA

Manufacturing Design Engineering Intern

May 2025 - May 2026

- Delivered \$2.8M+ in savings by leading DFM, tooling, and design improvements such as hot-runner conversion, CNC and polishing cycle-time reductions, physical vapor deposition (PVD) elimination, and a 71% raw-material cost cut.
- Raised assembly yield (65% → 99.7%) by adding computer-vision feedback and reducing tolerance stack in fixturing.
- Reduced assembly time 50% and cost 40% through design changes to qualify laser welding over mechanical fastening; cut CNC cycle time 260s, reduced part count, and coordinated validation across PD, EE, and FATP Ops.
- Pioneered PVD-free passivation surface finish for corrosion resistance and tunable color by leading R&D end to end; subject matter expert consultation, trial design, on-site vendor experiments, and the industrial-design review.
- Traced a critical strength failure to hydrogen embrittlement through root-cause analysis; >2x yield strength by implementing a bakeout; moved parts from failing to passing FEA requirements.

Eikon Therapeutics

Hayward, CA

Mechanical Engineering Intern

June 2024 - August 2024

- Increased laser precision 44.4% and cut alignment time 90% by building an automated optical test rig (Python controlled hardware, beam profiler, custom GUI) for single-molecule tracking microscopy.

Design Visionaries

San Jose, CA

Design Intern

May 2023 - August 2023

- Filed 4 USPTO provisional patents for small businesses by designing and prototyping medical and consumer devices in Siemens NX, validated with DFM analysis.

Research and Club Experience

Musculoskeletal Biomechanics Lab

New York, NY

Undergraduate Researcher

September 2024 - Present

- Built the anatomical CAD models used for printing and testing by other Columbia labs on the NITRO NOVA knee-replacement project; designed test fixtures and a LabVIEW DAQ system for implant validation.

Creative Machines Lab

New York, NY

Undergraduate Researcher

February 2024 - Present

- Redesigned a food 3D printer for a Michelin-star chef, converting the extrusion system from a syringe to a pressurized hopper-and-auger feed for larger volumes; built a Python G-code pipeline to improve print consistency.

Columbia Makerspace

New York, NY

Superuser Volunteer

January 2024 - Present

- Host weekly office hours and train students on fabrication equipment (laser cutter, 3D printer, waterjet, woodshop).

Engineering and Design Projects

Handheld Audiobook Player ([View Here](#))

- Audiobook player built off a raspberry pi and python featuring auto-transcription for quote capture, bookmarking, and speed control; hardware prototype completed using 3D-printing for enclosure design.

Pinball Machine

- Built an F1-themed pinball machine by opening up machines at a local pinball museum to reverse-engineering commercial components, designing a playfield in Fusion360, and integrating CNC-machined parts with electronics.

Multi Resistance Mechanism Foot Brace

- Led 6-person team to design and prototype a multi-resistance foot brace for foot drop patients, conducting user research with MS patients and validating with a working prototype.

Skills & Interests

Software: Siemens NX • Solidworks • Blender • LabVIEW • JMP • Python • C • Java • Raspberry Pi • Adobe Suite

Manufacturing: DFM • DOE • FMEA • Tolerance Analysis • Injection Molding • 3D Printing • CNC • Laser Cutting

Hobbies: Powerlifting • Djing • 3D Modeling • Reading • Board Games • Needlework • Book Binding • Making